Modeling Phase 1 Clinical Trial Designs

Evaluating the 3+3, mTPI, mTPI-2, and CRMs

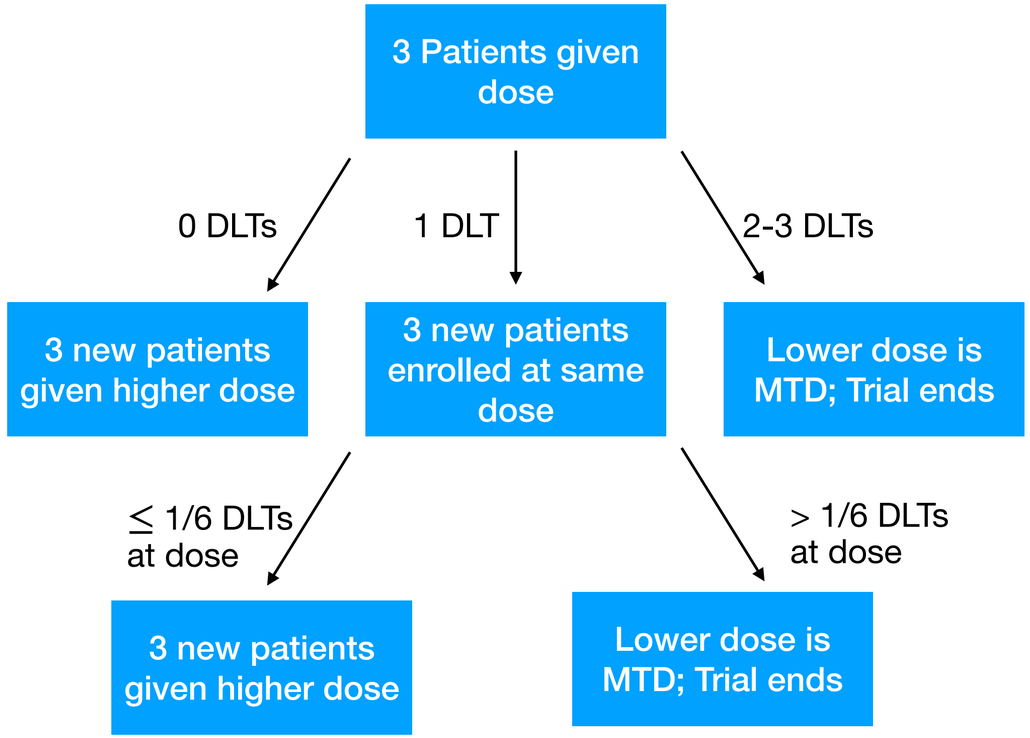
Jonas Schiel, February 28th, 2026

Phase 1 Clinical Trials

- Focus on finding dosing limits that optimize safety and efficacy of the treatment
- MTD
  - Maximum Tolerable Dose
  - Highest dose level that can be administered
- DLT
  - Dose Limiting Toxicity
  - When a patient experiences a level of toxicity that results in adverse reactions
  - Binary response; 0 = No DLT, 1 = DLT

3+3 Designs

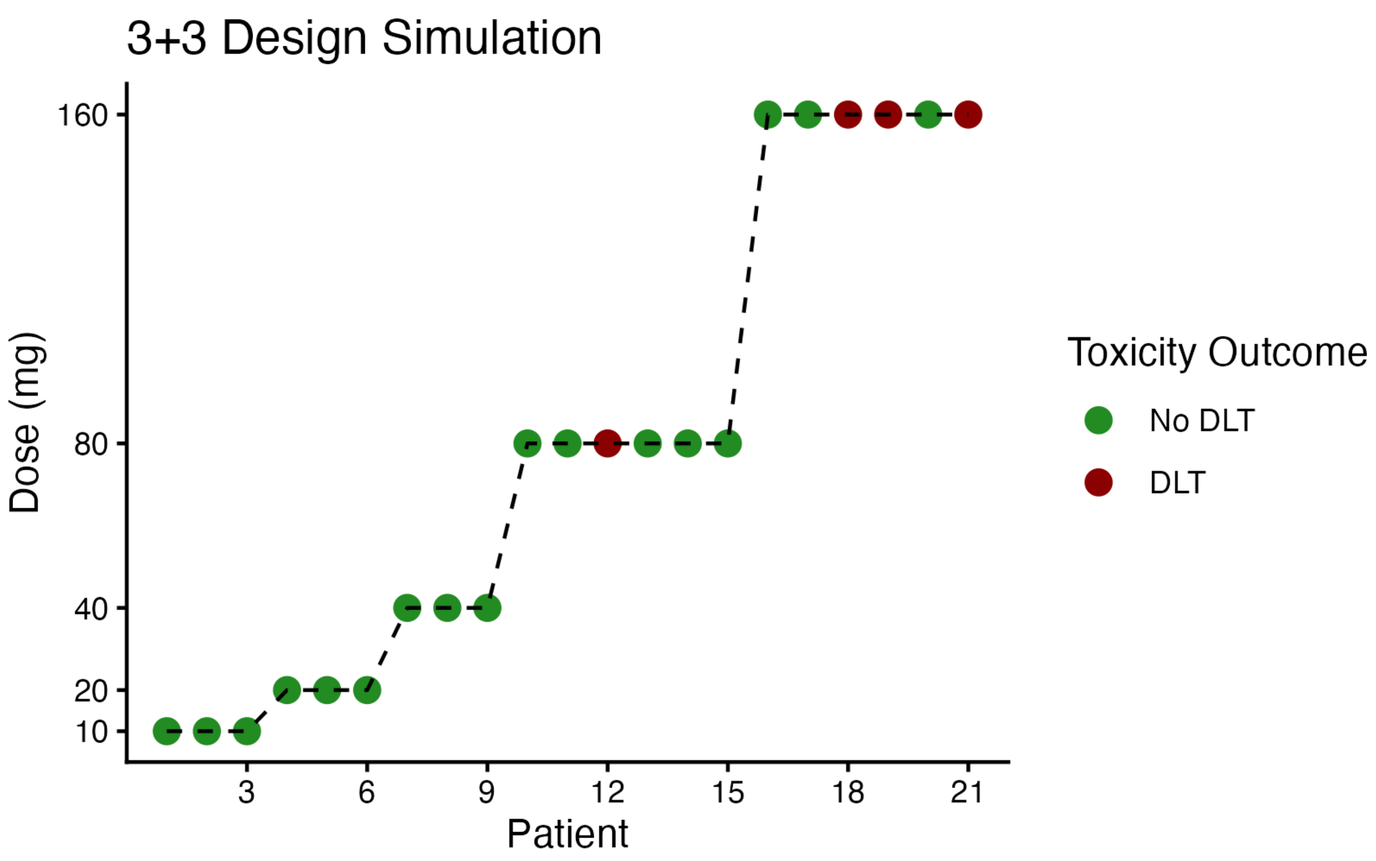
- Currently the most applied design method used in Phase 1 clinical trials
- “More than 95% of Phase 1 trials have been based on the 3+3 design” (Ji and Wang, 2013, p.1785)
- Treats cohorts of 3 patients at a time; Each dose is only tested a maximum of 6 times across 2 cohorts (variations exist)
- The model follows an algorithmic pattern, requiring no application of statistics or computer programming to determine decision making
- The model follows a rigid structure that can’t easily be altered for different target toxicities



DLT = Dose Limiting Toxicity

MTD = Maximum Tolerable Dose

3+3 Simulation



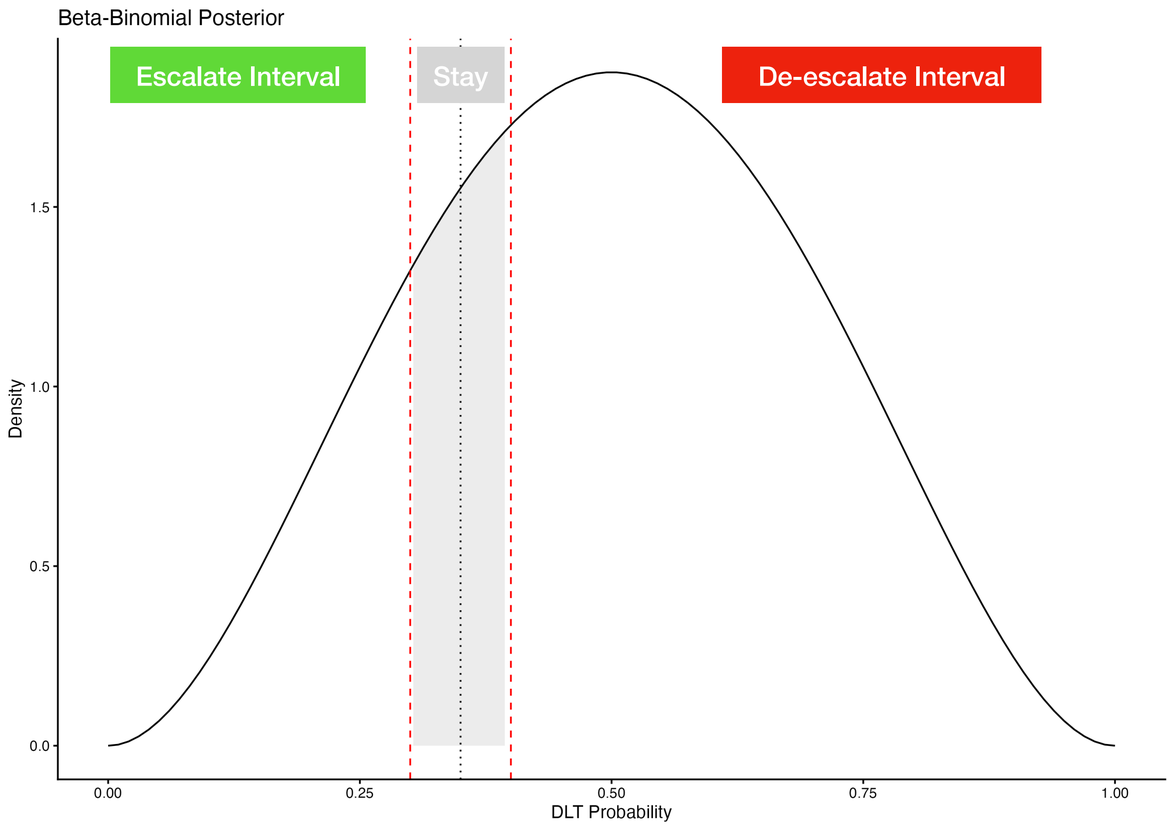
3+3 Dose Assumptions Used in R Simulation	
Dose Level (mg)	True DLT Probability
10	1%
20	5%
40	10%
80	25%
160	45%
320	55%

mTPI

Modified Toxicity Probability Interval

- Allows trials to admit one patient at a time
- Relies on a Beta-Binomial Posterior for decision making
- Often uses non-informative prior, Beta(1,1)
- The posterior probability of a patient experiencing a DLT is normalized across 3 intervals (Escalate, Stay, De-escalate)
- The design quantifies UPM for each interval and selects the interval with the maximum value

$$UPM_E = \frac{Pr(M_E \mid data)}{Length_E} \qquad UPM_S = \frac{Pr(M_S \mid data)}{Length_S} \qquad UPM_D = \frac{Pr(M_D \mid data)}{Length_D}$$

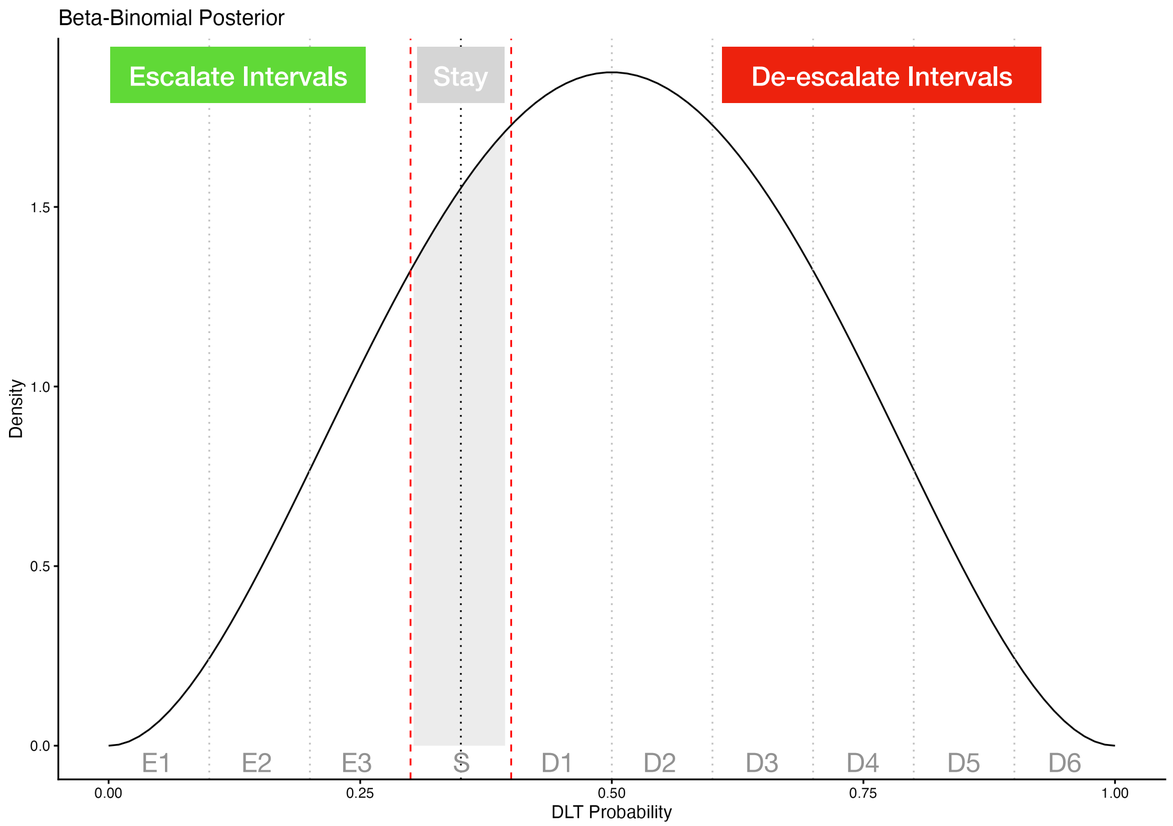


mTPI-2

Modified Toxicity Probability Interval

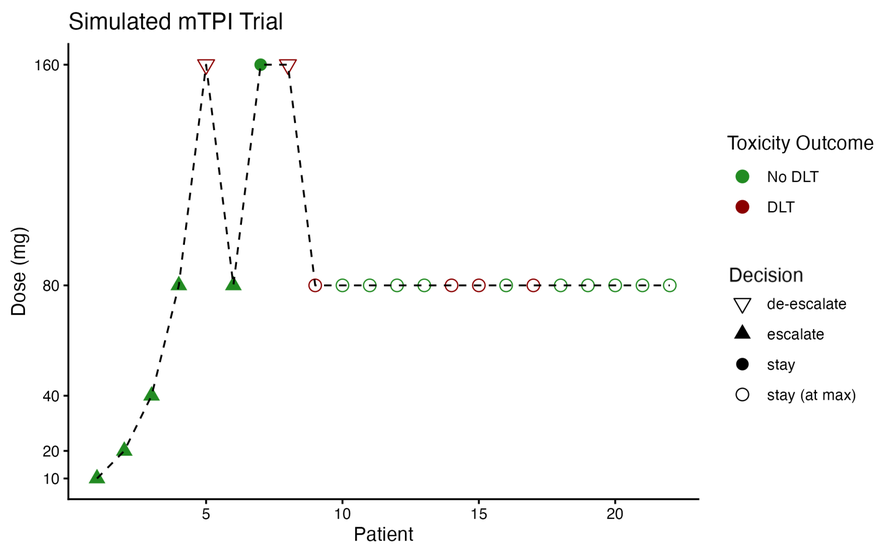
- mTPI model that uses rolling intervals for decision making
- Minimizes punishment on escalation and de-escalation intervals
- Doses are escalated and de-escalated more frequently as a result

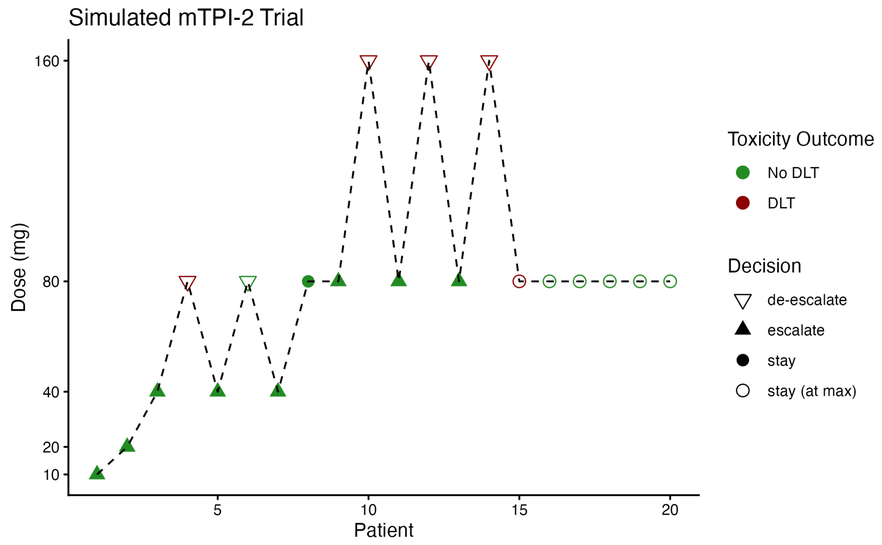
$$UPM_{D1} \frac{Pr(D_1 \mid data)}{Length_{D_1}}$$

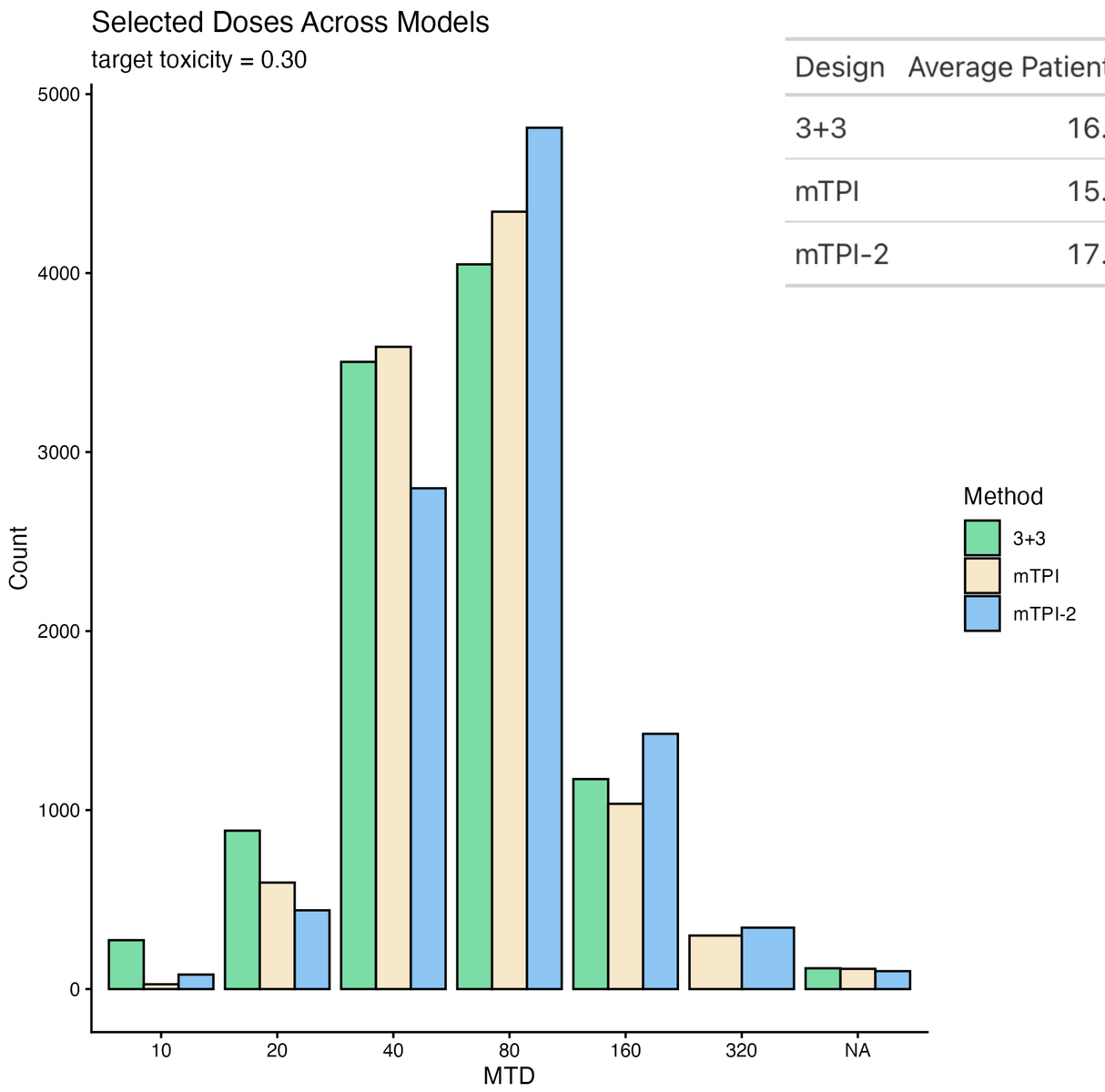


Simulating mTPI Trials

- Compare mTPI, mTPI-2, and 3+3 trial designs side by side
- Evaluate stopping rules and dose termination rules to ensure safe trials
- Stopping rules used in these trials:
  - Patients treated cannot exceed 30
  - If $Pr(DLT_j \mid data) \geq 0.65$, the dose is suspended from the trial
  - Trial ends when a 95% Credible Interval at tested dose falls below a width of 0.45







Design	Average Patients	Average Trial Toxicity	Median MTD	True MTD	Overdosing	Average DLTs
3+3	16.6	0.17	80	41.1%	11.6%	2.8
mTPI	15.8	0.26	80	44.1%	13.5%	4.1
mTPI-2	17.0	0.27	80	47.9%	17.8%	4.7

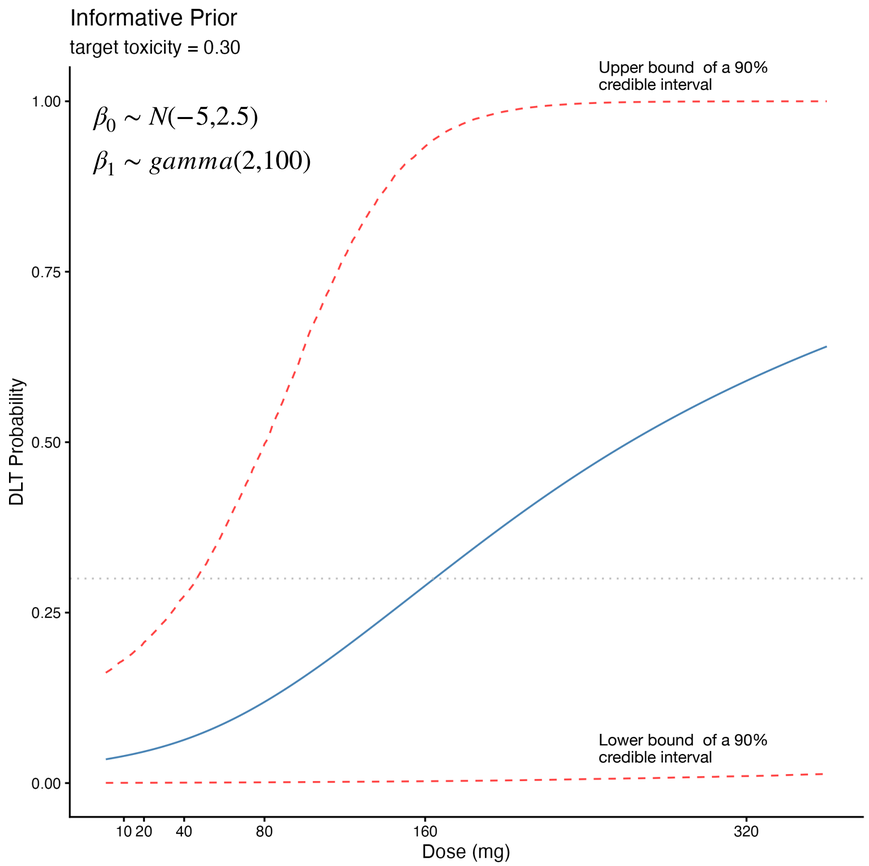
Dose Level (mg)	True DLT Probability
10	1%
20	5%
40	10%
80	25%
160	45%
320	55%

CRMs

Continuous Reassessment Models

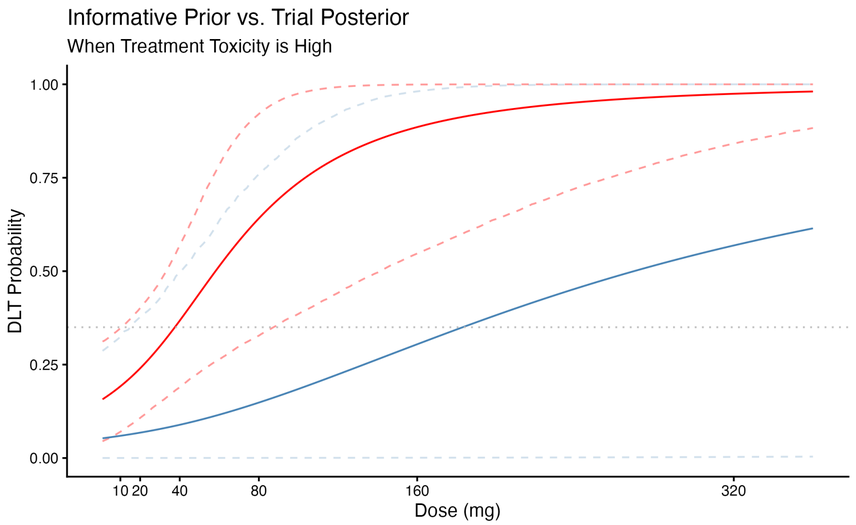
- Uses logistic regression to model the dose response
- Allows for information sharing between dose levels
    - Each observation helps to tune the model regardless of dose, typically resulting in faster decision making
    - Uses one informative prior for the dose response
- Assumes the response is monotonic, this assumption is met by ensuring β_1 is positive
- For my design β_0 was modeled using normal distribution, β_1 was modeled using gamma distribution

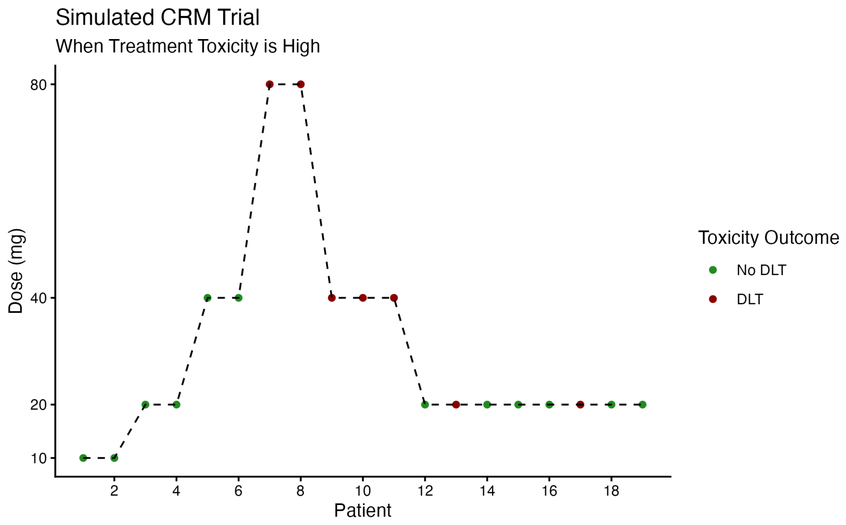
$$Pr(DLT \mid d) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot d)}}$$

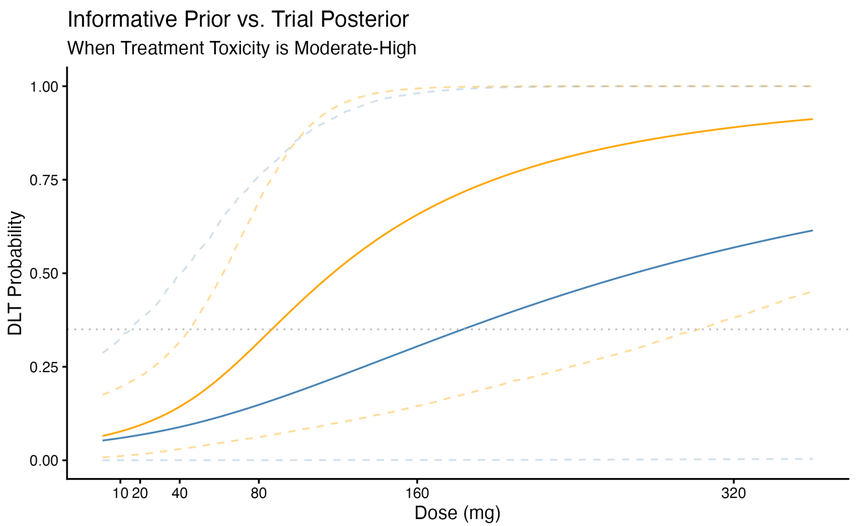


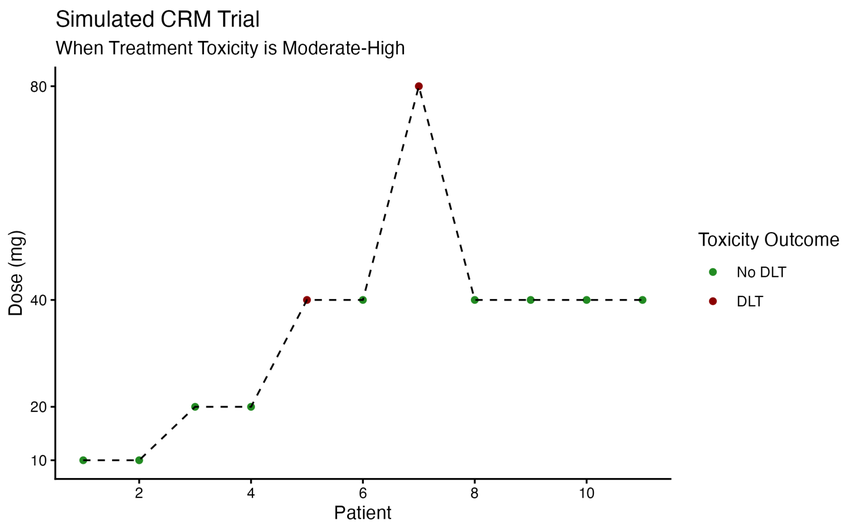
Evaluating CRM

under high dose toxicities



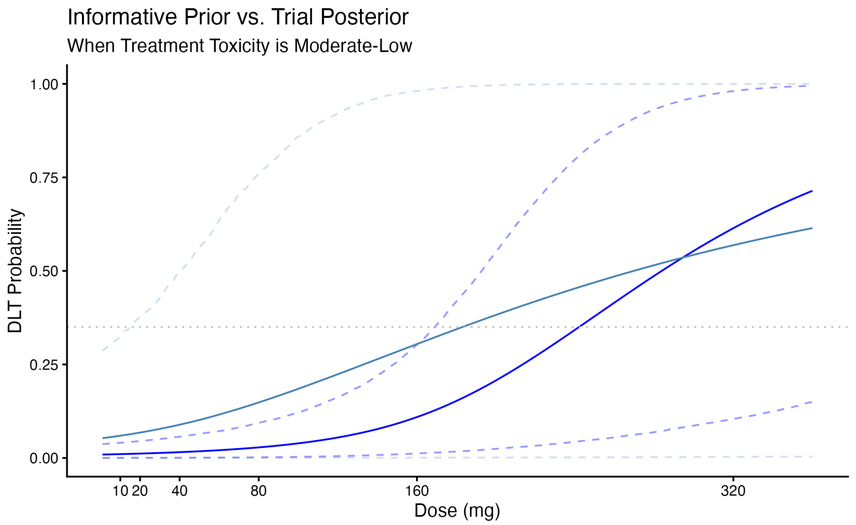


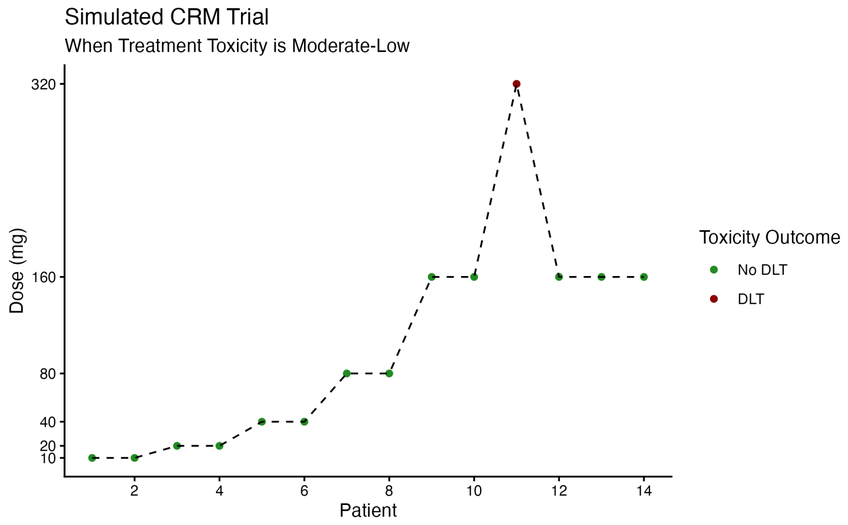


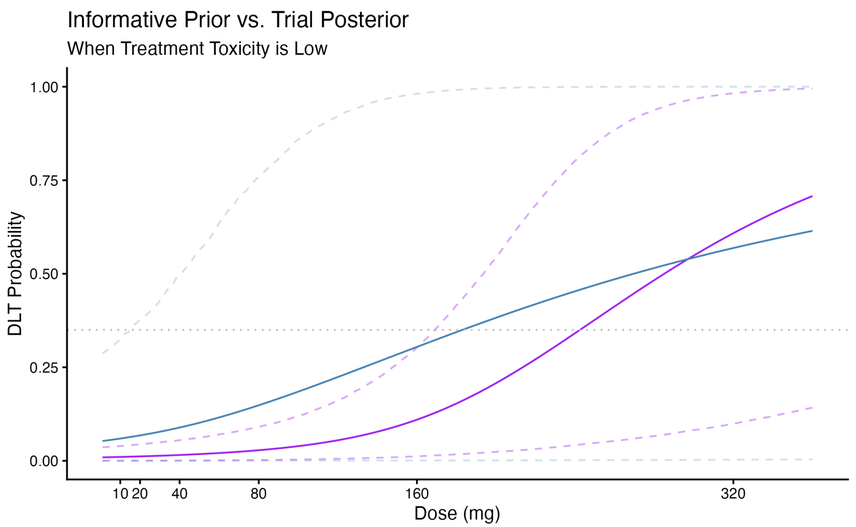


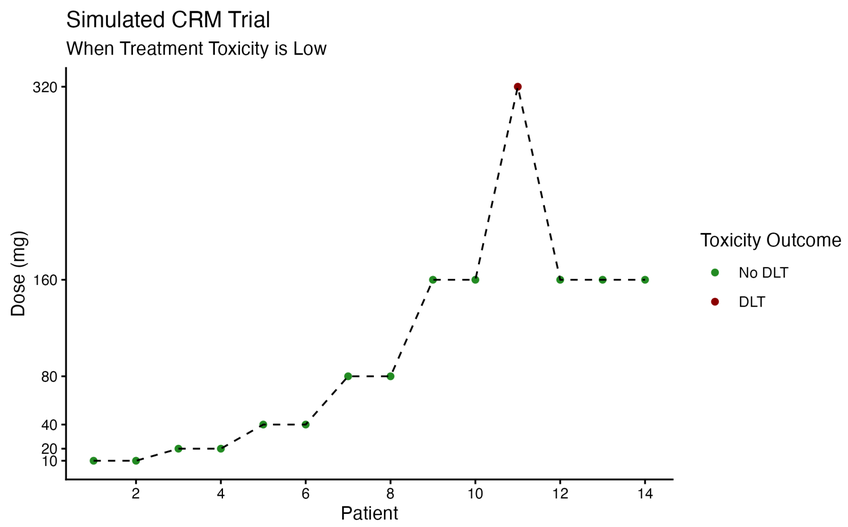
Evaluating CRM

under low dose toxicities









Future Research with CRMs

- Simulate CRM models repeatedly to generate metrics such as average patients per trial, proportion of trials with correct dose selection
- Determine trial rules that result in less patients required and improve accuracy in dose selection
- Evaluate the possibility of stopping at the credible interval width of dose range at the target toxicity opposed to the DLT probability at the last tested dose
    - Improve interpretability of design outcomes
    - More clearly defines the MTD, potentially limiting selection of excessively toxic doses